15
Application Areas of Computer Vision and AI in Intelligent Automation Systems

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Abstract

The combination of artificial intelligence (AI) and computer vision (CV) has led to tremendous growth in industries such as manufacturing, healthcare, agriculture, transportation, and sports. BMW applications show how robots guided by vision can solve 97% of problems accurately, so product quality control drives it forward. AI-based medical imaging is expected to have a significant impact on healthcare. These technologies have led to streamlined assembly lines, and integrated analytics in manufacturing processes to increase efficiency and reduce costs. In healthcare, AI-based medical imaging is expected to play a major role in accurate diagnosis and personalized patient care. With the global market estimated to reach $45.13 billion by 2027, the safe, autonomous vehicle market will reach about $556.67 billion by 2026. Moreover, the impact of AI and CV technology is expected to affect the sports analytics market, which is estimated to reach $4.62 billion by 2025. Essentially, AI and CV are enabling intelligent automation and restructuring of businesses around the world.

Keywords: Computer vision, AI technologies, manufacturing, healthcare, agriculture, transportation, sports, performance analysis

15.1 Introduction

Today, the emphasis is placed on systems with technological advances that combine “artificial intelligence and intelligent systems.” But one of the last important technologies he acquired was AI technology. Since real-life situations are dynamic and diverse, often with a large amount of in-

formation [1], the development of a robust AI model is essential. Artificial neural networks can learn from data to generate recommendations and help the decision-maker make decisions.

This sounds like "smart thinking" requires efficiencies comparable to, say, Chatbot or intelligent business assistants [2]. The text intelligence component refers to text analytics, whether written or spoken, which enables companies to gain expertise in text recognition, language translation, and text interpretation in multiple languages [3]. In the fourth industrial revolution, humanity is on the brink of unprecedented technological advancement, transforming the lives of businesses, societies, and individuals on an unprecedented scale, artificial intelligence (AI), automation, and communication are communicating an exciting new era of innovation and change. As automation becomes more embedded in everyday life, the need for cognitive processes capable of processing, analyzing, and responding to large amounts of data has increased, and where AI is becoming a key enabling force in a wide range of industries and applications.

With the help of deep learning (DL) and machine learning (ML) techniques, machines can understand and perceive the visual world like human emotions [5].

An important part of AI and computer vision (CV) has emerged as a transformational project changing all industries globally. This chapter begins with studying the basic methods applications, and the profound impact of computers in various industries. CV systems can mimic human visions to detect objects, interpret images, and even understand complex spatial relationships. As AI has become more common, many application areas have seen a dramatic shift in how tasks are performed, decisions are made and results are achieved Quality control and predictive maintenance in robotic manufacturing inventory management are traditional approaches of AI technology that are also changing just some improve productivity and encourage innovation [6] CV, ML, DL, Internet of Things (IoT), and robotics are enabled by many important technologies that underpin intelligent systems. In healthcare, CV and AI are revolutionizing diagnostic and therapeutic medical imaging. It facilitates the development of precision therapy using techniques [7]. Similarly, monitoring patient vital signs and activity levels through wearable sensors, IoT devices, ML and DL models have become personalized healthcare solutions. Apart from medicine, AI is also gaining ground impacts in other areas such as chemical manufacturing, crop management, and environmental protection. ML algorithms measure cells and monitor environmental contaminants, while DL models use chemical databases to promote drug discovery [8].

15.2 Advanced Techniques in CV and AI for IAS

AI is the driving solution such as customer behavior analytics, security systems, and navigation planning. Autonomous assembly vehicles optimize logistics and strengthen safety measures, and facial recognition technology enables appropriate marketing strategies. They breed intelligence-developed products to stimulate human intelligence with the help of the computer program ML. DL, a subgroup of ML, revolutionized the field by introducing multidimensional neurons. As a result, information can be presented in abstract and complex ways. The evolution from early AI concepts to current DL models is illustrated in Figure 15.1.

15.2.1 The Power of Deep Learning (DL)

A subclass of DL, ML uses multilayer neural networks to extract complex features from complex data, enabling higher levels of abstraction and representation. This approach has led to breakthroughs such as CV, NLP, and speech recognition that have helped drive growth in various industries. In particular, DL using convolutional neural network (CNN) algorithms has become increasingly important in revolutionizing computers, enabling them to learn categorical data directly from raw pixels [10]. CNN architectures have been successful in object recognition, image segmentation, and classification. AlexNet [11], VGGNet [12], and GoogLeNet [13] are examples of such models that highlight the capabilities of CNN in complex image recognition tasks.

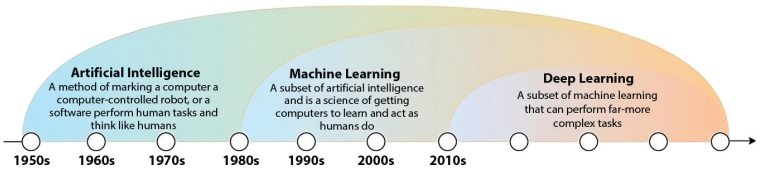


Figure 15.1 Evolution of AI, ML, and DL technologies.

The concept of a recurrent neural network (RNN) revolves around the processing of sequential data, maintaining its internal state, and acting as memory. This processing can preserve patterns and temporal relationships in sequential data. RNN finds applications in time series analysis, speech recognition, and natural language processing (NLP). Sophisticated paradigms such as long-term and short-term memory (LSTM) [14] and twoway LSTM (BLSTM) [15] have shown promising results in tasks such as emotion analysis, speech interpretation, and speech recognition.

An inception network is a system that uses multilevel processing to capture objects at multiple levels of abstraction. ResNet adopts salt switch interfaces to mitigate the stray flow problem, which facilitates the training of highly robust networks [16]. DenseNet facilitates efficient feature reuse and concise model representation by connecting each layer to all other layers through skip connections [17]. Meanwhile, given the increasing use of mobile-embedded devices for distributed computing, the mobile net has been developed to reduce computational complexity while maintaining high performance [18].

15.2.2 Exploring IA Methods and Solutions for Future Evolution

Now a need for flexibility through advanced technology to automate all processes. AI focuses on and applies key concepts. This is important for technology development and decision-making in information technology. The aim is to provide critical guidance for those interested in CV and AI research and development in IAS-based innovations across sectors [19]. Table 15.1 provides an overview of model performance and includes an example of its use. The following important areas are crucial for the future development of CV and AI.

- Finding Research Areas: This section helps identify research areas related to automation, cognitive engineering, and real-time decision-making by focusing on current needs, e.g., automated quality control systems using AI to improve manufacturing efficiency and reduce manufacturing errors.

- AI Categories: This includes many AI teams such as analytical, functional, reactive, text, and visual compression, ie NLP algorithms enable machines to understand and produce text similar to that written by humans transforming information search and communication [20].
- Emphasizing the Role of Advanced AI: Highlight the critical importance of advanced AI in understanding and decision-making, i.e., in the fourth industrial revolution. Companies can anticipate market trends and optimize distribution thanks to AI-powered predictive analytics equipment.
- In-Depth Analysis of AI Techniques: Used to develop intelligent automation models, these techniques improve analytical capabilities and real-world applications, ie. financial transactions using ML algorithms such as SVMs to identify unique fraud detection patterns.

Table 15.1 Model performance and application examples by reference.

Year	Model	Reference	Example
2012	AlexNet [11]	Classification of ImageNet with deep CNN	Image classification (IC), object detection (OD)
2015	VGGNet [12]	Very deep CNN for large-scale image recognition	IC, feature extraction
2014	GoogLeNet [13]	Going deeper with convolutions	OD, IC
2014	Inception V1 [21]	Re-thinking the inception architecture for computer vision	OD, IC
2016	ResNet [22]	Deep residual learning for image recognition	OD, IC
2016	DenseNet [23]	Densely connected convolutional networks	OD, IC
2017	Inception V3 [24]	Rethinking the inception architecture for computer vision	OD, IC
2017	LSTM [14]	Long short-term memory	NLP, time series analysis
2017	BLSTM [15]	Framewise phoneme classification with bidirectional LSTM and other NN architectures	Speech recognition, sequence prediction
2017	MobileNets [25]	MobileNets: Efficient CNN for mobile vision applications	Mobile and embedded image recognition

This paragraph illustrates the evolution of DL frameworks, which by 2020 will introduce image classification, object recognition, and sequence mapping services featuring advances in object detection, image segmentation, and sequence mapping.

- Leveraging AI-Based Solutions: Researchers and developers can learn AI technologies by engaging AI-based solutions in various practical areas. For example, AI-based diagnostic systems help doctors diagnose diseases more accurately.
- Identifying Research Problems: Identifying and solving research problems helps guide future research design, process improvements, and improvement steps, i.e., address issues like data privacy. Algorithmic bias is necessary to ensure ethical and unbiased applications and use of intelligent technology.

15.3 Why We Use AI in Research and Services Today

In today's controversial world, the main focus is on getting people interested in CVs and AI techniques that can be applied in various industries. The AI Revolution delves into the ground-breaking technologies of Industry 4.0, focusing on integrating them with high-performance machines and DL techniques. [26] explore research methodologies and practical applications in this growing field.

The impact of AI can improve existing products and services by increasing performance, reliability, and safety. Let's take the field of AI, where data visualization using AI prevents accidents and improves navigation. This is followed by increased investment in AI solutions. Major players in various industries are racing to develop new AI projects. This adoption craze is not confined to one place; it permeates offices, hospitals, entire cities, and the digital world. AI is changing how problems are solved and solutions provided, making its integration into modern operations an important consideration [27].

The primary goal of AI is to understand and perform intelligent tasks, from problem-solving to adapting to dynamic situations, so AI remains a key area of development. Continuous research and development highlight the importance of experimentation. However, due to ever-changing real-world scenarios and vast amounts of information, building effective AI models remains a challenging endeavor. Intelligent automation (IA) involve digital processes and user experience, and understanding the intricate details of the process is paramount. Different aspects of the IA are highlighted through interesting descriptions such as Figure 15.2, which extends from the web design to the content. Such visual cues are a valuable tool for understanding complex IA processes, ultimately increasing user satisfaction and usage of digital service interfaces.

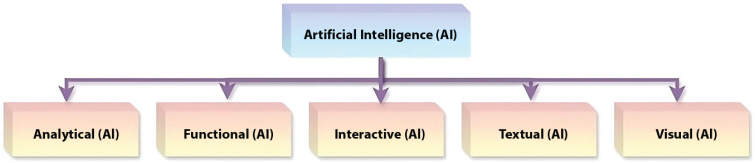


Figure 15.2 Highlighting several forms of IA.

15.4 The Association Across AI, ML, and DL

The amalgamation of AI, ML, and DL is a part of formulating technological progress. Figure 15.3 shows the interaction between ML and DL in the AI domain. DL is specifically identified as a subcategory embedded in ML and AI frameworks, highlighting its important role in a broad range of AI applications, as in Figure 15.3. AI technologies often influence human behavior domains alongside machine or system intelligence, while ML, empirical, or knowledge-based predictive create models automatically [28]. DL uses knowledge-based learning techniques using multilayer neural networks and advanced processing tools to generate computer models [29]. The term “deep” in DL emphasizes the many layers or steps involved in processing data to achieve model-based insights [30]. This complex relationship between AI, ML, and DL forms the basis for AI development and innovation, with any improvements made by the industry contributing to the recovery of the environment [31].

Different ML techniques represent a particular subject; depending on its principles and pedagogical capabilities. One common approach is supervised learning, where a specific goal is defined as achievable based on inputs. These “task-based methods” use labeled data to classify data or train algorithms to predict outcomes, similar to how spam detection works. Supervised studies go through classification for quantification and regression analysis for magnitude. Decision trees *viz.* ID3, C4.5, and CART [32] are often used. Cluster learning methods such as RF (random forest), LR (linear regression), and VC (vector regression) have been widely used to solve many problems [33].

The combination of IA (intelligent automation), ML, and DL is key to shaping the predictive modeling landscape. The general structure of ML prediction models consists of supervised learning tasks in the generation and validation phases designed for a specific problem [34]. On the other hand, unsupervised learning is often referred to as a “data-driven method.” It includes tasks such as classification, visualization, dimensionality reduction, association rule detection, and anomaly detection [33]. Various methods *viz*. integration of many unsupervised learning activities with learning algorithms such as K-Means, K-Medoids, CLARA, DBSCAN, AIS, Apriori, FP-Tree, and selection methods such as Pearson correlation and principal component analysis (PCA) [35]. This synergy between IA, ML, and DL highlights their ongoing synergy in promoting innovation and solving problems in different sectors.

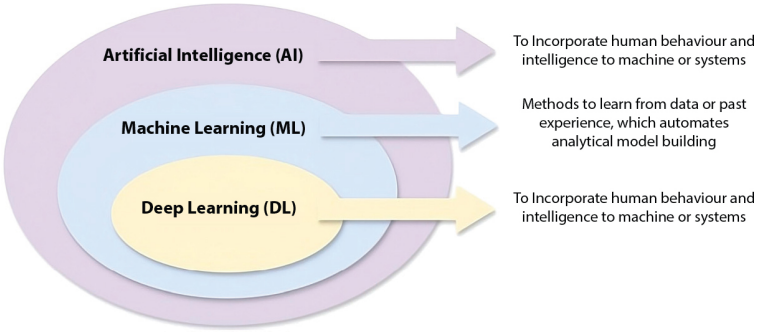


Figure 15.3 AI, ML, and DL technologies.

The combination of AI, ML, and DL is connected to solve complex tasks. One such approach, semi-supervised learning, uses unlabeled and labeled data to generate models by combining unsupervised and supervised methods [36]. This hybrid approach proves especially useful when automatically labeled data is required, so it is useful in situations such as Internet content or text classification [37]. Desirable behavior is reinforced, and undesirable conduct is discouraged [38]. In a reinforced learning environment, participants learn through experimentation and interaction with their environment. Common learning methods include Q-learning, Monte Carlo, and deep-Q networks [39].

15.5 Exploring Deep Learning and Neural Systems

DL is a popular AI technique that relies heavily on artificial neural networks (ANNs) [40]. The ability to retrieve data from a variety of sources has made it a leader in modern computers. In a deep NN, there are many hidden layers such as input and output layers. Supervised learning includes algorithms that use labeled data, which facilitates tasks such as classification and regression [41]. In contrast, unsupervised learning has models that extract patterns from unwritten data and can be used in clustering, association discovery, etc. [42]. Semi-supervised learning tells stories two together to create models, which is useful when labeled information is limited. On the other hand, reinforcement learning revolves around model learning with environmental interactions, rewards, or punishments acquired based on one's actions [38]. These different learning styles illustrate the multidimensional nature of AI and its potential applications in various fields are presented in Table 15.2.

Table 15.2 Comparison of learning approaches and tasks in ML.

Learning type	Model building	Tasks	Examples
Supervised [41]	Learning from labeled data	Regression and classification	Image classification, stock price prediction
Unsupervised [42]	Learning from unlabeled data	Clustering and associations	K-means clustering, market basket analysis
Semi-supervised [43]	Use both types of data labeled and unlabeled	Dimensionality reduction and classification	Self-training classifiers, autoencoders
Reinforcement [38]	Models are based on reward or penalty	Control	Reinforcement learning for game-playing

15.5.1 Semi-Supervised Learning

It is integrating labeled and unlabeled data, especially in cases where labels are scarce or expensive. Self-learning classifiers start with labeled data and refine the model iteratively with unlabeled data, leading to increased performance over time [43]. A NN auto-encoder is often used for dimension reduction in semi-supervised learning environments.

15.5.2 Reinforcement Learning (RL)

The concept of RL focuses on exemplary learning through interaction with the environment, where rewards or punishments are given based on behavior. This approach is often used in applications that are used to maximize cumulative utility [38]. For instance, in a reinforcement learning game, agents learn the best strategies through trial and error to earn or win more points and win opportunities. RL algorithms such as deep-Q networks and Q-learning have been successfully applied in various fields [39] including robotics and sports.

15.6 Delving into Deep Neural Networks' Learning Approaches

The global topology of deep neural networks (DNNs) with $N \geq 2$ layers is quite different from that of SNNs with $N \geq 1$ layers. DNNs have been widely used in supervised learning tasks or discrimination and are aimed at providing discrimination tasks in various applications. Various architectures such as multilayer perceptron (MLP), CNN, RNN, and their variants have been developed to solve specific classification problems [11]. In ad-

dition, deep correlations have been used in unsupervised or generative learning environments to capture high-level correlation features for model analysis or combination [33]. These techniques reduce reliability when they are studying specific information processing, commonly used in information systems description tasks Boltzmann machine differences [44] such as RBM, self-organizing maps (SOMs), and deep belief networks (DBNs) [49]. Moreover, learning hybrid approaches have resulted in combining discriminative and reproductive models and taking advantage of the strengths of both models [45]. This hybrid model combines elements of supervised and unsupervised learning, leading to better results in solving real problems [41].

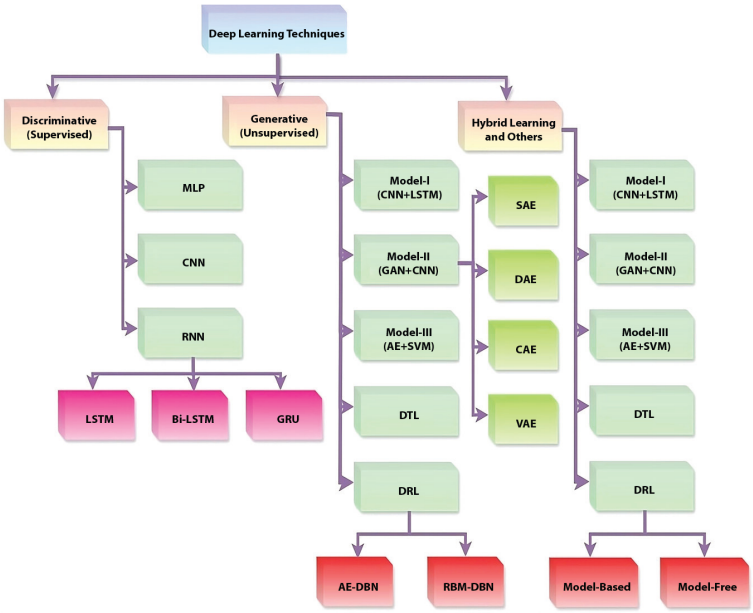


Figure 15.4 DL taxonomy as supervised, unsupervised, and hybrid learning.

Figure 15.4 illustrates the classification of the DL model into supervised and unsupervised hybrid learning, which provides a systematic framework for understanding and implementing different DL frameworks [42].

15.6.1 Unveiling Deep Neural Networks Learning and Analytical Methods

Research on deep neural network (DNN) learning techniques reveals a versatile approach to data mining, data discovery, and advanced analysis [46]. Figure 15.5 shows a comparative analysis of data analysis and data science components. Table 15.3 classifies the analytical methods and shows their application to the model-building and decision-making processes.

Data science is a multifaceted field that deals with fundamental questions; “What happened?”, “What caused it?”, “What’s next?”, and “What kind of work should be done?”— These questions form the basis of four main categories of analysis: descriptive, analytical, predictive, and prescriptive, each of which contributes to the development of a corresponding data model.

Advanced predictive analytics uses historical data to predict future events with high confidence by simply asking the question “What will happen in the future?” Such research not only examines the past but also uses historical data to predict and predict future behavior. For example, authors of [47] implemented a decision support system using data mining

and financial information in Arab universities and proposed a data mining approach to improve online business competitiveness, while [48] used data mining to study threats in the cyber domain. [49] showed how diagnostic methods and automated analysis can be used in the gas industry to reconstruct production events. Furthermore, the authors of [50] developed a standardized research program to improve health counseling processes through automated learning algorithms and programming rules. This study has demonstrated the different applications and benefits of data analysis techniques in various industries.

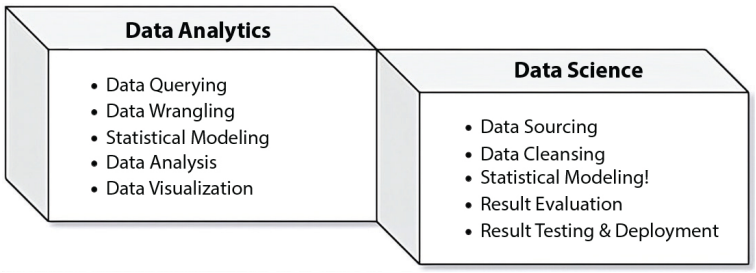


Figure 15.5 Comparing components of data analytics vs. data science.

Table 15.3 Classification of analytical methods.

Analytical methods	Data-driven model building	Description	Examples
Descriptive analytics [1]	What happened in the past?	Examines historical data to understand company evolution.	Summarizing past events like sales, social media usage, business data, reporting general trends, etc.
Diagnostic analytics [2]	Why did it happen?	Analyzes information to determine the root cause of events.	Identifying anomalies and determining causal relationships, investigating business loss, medication influences, etc.
Predictive analytics [3]	What will happen in the future?	Utilizes historical data to forecast future events with high certainty.	Predicting customer preferences, predicting staff and resource needs, etc.
Prescriptive analytics [4]	What action should be taken?	Recommends action plans based on data to optimize revenue and profitability.	Improving business management, enhancing patient care and healthcare administration, maintaining procedures, etc.

15.7 Rule-Based Modeling: A Cornerstone of AI Development

Rule-based systems are the cornerstone of data processing and enable data to be stored and exchanged in a manner that allows for appropriate

interpretation. In general, programs formulated as IF-THEN statements excel in capturing human knowledge-like decision-making processes [51]. They are characterized by adaptability, which allows for rapid adaptation by incorporating new rules based on emerging knowledge or trends [52]. Historically, such systems relied on manual codes but advances in learning techniques have led to a shift to automated methods [53]. Using classification techniques such as decision trees, associative learning methods, data mining techniques, and data mining algorithms, researchers have developed sophisticated models capable of powering complex [54]. Retained modeling in AI development simulations and decisions provides a robust framework for solving real-world challenges in various sectors [55]. Table 15.4 provides insights about rules-based frameworks in AI applications.

Table 15.4 Insights into rule-based systems in AI applications.

Citation	Facts/figures	Examples in AI/CV in IAS
[51]	Rule-based systems store and modify data in meaningful ways using IF-THEN statements.	Rule-based models exhibit decision-making processes akin to human expertise in intelligent automation systems (IAS).
[52]	Adaptability allows swift adjustments by incorporating new rules based on expert knowledge or emerging trends.	Automation of rule-based systems in IAS enables rapid adaptation to changing environments or requirements.
[53]	Advancements in learning strategies have led to a shift toward automated rule-based approaches.	Integration of data mining algorithms enhances the efficiency and accuracy of rule-based models in IAS.
[54]	Rule-based models demonstrate efficacy in diverse applications, from adaptive mobile systems to risk assessment in healthcare and air traffic control.	Examples include adaptive mobile systems for context-based decision-making, risk assessment for disease identification, and early warning systems for air traffic control in IAS.
[55]	Rule mining techniques identify risk factors for adverse diseases.	Rule-based models aid in identifying risk factors and patterns for proactive decision-making in healthcare within IAS.
[56]	Rule-based models enable risk prediction and early warning systems for air traffic controllers.	Utilization of rule-based models enhances safety measures and decision-making processes in air traffic control systems within IAS.

15.8 The Role of Fuzzy Logic and Distributed Logic in AI

Fuzzy logic is a mathematical method that extends traditional logic by considering degrees of truth from 0.0 to 1.0 based on uncertainty approximation [57]. Unlike traditional logic, which operates on binary truth values, fuzzy logic works on a continuum. Distributed logic systems enable fine-grained representation of uncertain and fuzzy data, as exemplified in Figure 15.6. They explore various concepts like fuzzy inference systems, membership functions, fuzzy sets, fuzzy logic operators, and their applications in real-world scenarios.

These algorithms handle irregular and noisy data well suited for uncertain real-world applications [58]. The authors of [59] used rational multi-interpretation classification to optimize cardiovascular disease diagnosis, while [60] proposed an AI system based on fuzzy logic for IoT-compatible environment monitoring [61] for using anomaly detection by distributed logic for network security and designing distributed products. Additionally, rules of thumb, contexts such as otologies, and rule-based systems [62] play an important role in decision-making in intellectual property [63]. Formal concepts of the knowledge domain and ontologies facilitate interactions between humans and machines that improve understanding and reasoning [64]. Moreover, probabilistic reasoning methods that combine logical and probabilistic reasoning provide effective solutions for dealing with uncertainty in AI [65, 66], encouraging fuzzy thinking, distributed thinking, and integrated reference strategies. Fuzzy thinking, distributed reasoning, and reference strategies promote intelligent decision-making in complex environments, as described in Table 15.5.

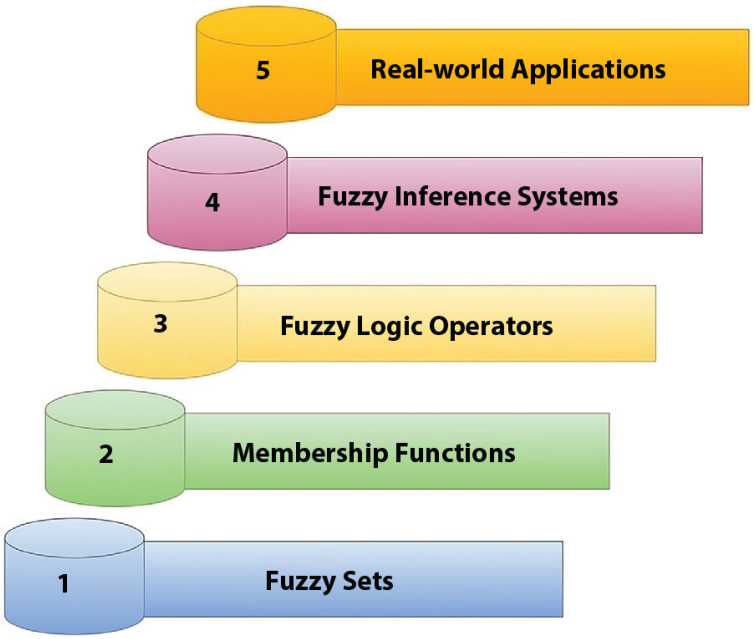


Figure 15.6 Illustrating the integration of fuzzy logic and distributed logic within AI.

15.9 AI and CV Technologies for Advancing Manufacturing Industries

Manufacturing companies use advanced technology to increase productivity and efficiency in various industries [72]. Predictive maintenance is needed to reduce downtime by anticipating device failures using IoT sensors and data analytics [73]. Automated robotic assembly systems combine robotics, CV, and AI systems for accuracy and efficiency in assembly processes [74]. Supply chain management is streamlined through automated inspections using RFID technology, CV, AI algorithms, and other tools to improve supply chain efficiency [75]. Table 15.6 presents numerous application areas within the manufacturing industry.

15.10 AI and CV Revolutionizing Healthcare Innovations

Many applications in healthcare technology use advanced techniques and data sets to improve disease diagnosis, patient care, and drug develop-

ment. CNN is applied to DICOM images in medical image analysis, ensuring high accuracy, sensitivity, and specificity [76]. Transfer learning methods applied to MR and CT scans provide better F1 scores, accuracy, and recall [77], while generative adversary network (GAN) improves X-ray image analysis through AUC and Dice coefficients high reach [78]. The use of wearable sensors and IoT devices enables monitoring of patient vital signs and activity levels, resulting in more accurate health information with lower error rates [79, 80]. ML models ensure continuous monitoring of data with high sensitivity and specificity [81]. DL patterns from drug databases contribute to drug discovery and development, resulting in impressive ROC AUC and F1 scores [82]. Reinforcement learning methods optimize the molecular structure to increase binding affinity and precision [83], while NLP extracts knowledge from the biomedical literature through accuracy, recall, and F-measurement [84]. Table 15.7 demonstrates the transformational impact of AI technology on healthcare.

Table 15.5 AI and knowledge representation in diverse domains.

Citation	Facts & figures	Examples
[59]	Logical ambiguity classification optimizes cardiovascular disease diagnosis.	Reddy *et al.* optimized cardiovascular disease diagnosis using a logical ambiguity classification.
[60]	Fuzzy logic–based AI systems enable IoT-enabled environmental monitoring.	Implemented a fuzzy logic–based AI system for IoT-enabled environmental monitoring.
[61]	Anomaly detection using distributed logic enhances network security.	Novaes *et al.* proposed anomaly detection using distributed logic in network security.
[62]	Distributed associative rule mining tailors product design to consumer preferences.	Kang *et al.* employed distributed associative rule mining for product design tailored to consumer preferences.
[64]	Knowledge representation, including ontologies, facilitates communication between humans and machines.	Li *et al.* presented ontologies, formalized conceptualizations of knowledge domains, and enhanced understanding and reasoning capabilities.
[66]	AI-based traffic monitoring relies on ontologies for intelligent decision-making.	Goel *et al.* provided a framework for intelligent traffic monitoring that is based on ontology and context.
[67]	AI control systems in marine exploitation leverage ontologies for intelligent decision-making.	Neupane *et al.* proposed a system of artificial intelligence control and intelligence based on ontology for intelligent marine exploitations.
[68]	Ontological data access control in cloud computing ensures secure knowledge management.	Kiran *et al.* presented a secure method and way to control access to ontological data in the context of computing in the cloud.

Citation	Facts & figures	Examples
[69]	Ontological cybernetic alarm systems manage cybersecurity vulnerabilities.	Merah *et al.* proposed a conceptual ontology and cybernetic alarm system for managing cybersecurity vulnerabilities.

Table 15.6 Application areas within the manufacturing industry.

Application area	Description	Key technologies
Quality inspection [70]	Automated inspection of products to ensure adherence to quality standards and detect defects.	Computer vision, machine learning, robotics
Predictive maintenance [71]	Predicting equipment failures to minimize downtime and optimize maintenance.	Internet of Things (IoT), data analytics
Robotic assembly [72]	Automated assembly of products using robotic systems to improve efficiency and precision.	Robotics, computer vision, AI planning
Inventory management [73]	Automated tracking and management of inventory levels to optimize supply chain operations.	RFID technology, computer vision, AI algorithms
Process optimization [74]	Analyzing production processes to identify inefficiencies and optimize resource utilization.	Data analytics, machine learning, optimization
Supply chain automation [75]	Automation of supply chain processes including order processing, logistics, and warehousing.	Robotics, IoT, blockchain technology

15.11 Innovative Solutions for Agriculture and Environment

The new approaches using AI and CV technologies in agricultural and environmental management are transforming traditional practices. Crop management and crop prediction will benefit from DL models using UAVs by analyzing satellite images [85]. ML algorithms that use knowledge from crop databases contribute to accurate crop growth prediction [86]. Remote sensing of hyper-spectral image data enables accurate crop health assessment [87]. Furthermore, pest detection methods and target-

ed pesticides use CV to identify and manage pests' view of the fields [88]. Similarly, IoT sensors facilitate real-time pest monitoring and alarm systems using field sensor data [89]. Furthermore, DL makes better use of historical outbreak data [90]. In environmental monitoring, CNNs are used to detect pollution from satellite images of the environment [91]. Sensor networks provide information for water quality monitoring in rivers and lakes, such as [92], and geographic information systems (GIS) to support habitat conservation by using environmental data [93]. These developments highlight the transformational impact of AI and CV on enhancing agricultural practices and environmental protection efforts. Table 15.8 presents application areas, methods, and datasets as the use is indicated.

Table 15.7 Application areas in healthcare technology.

Application	Method	Dataset	Performance metrics	Reference
Medical image analysis for diagnosis and treatment planning	Convolutional neural networks (CNN)	DICOM images	Accuracy, sensitivity, specificity	[76]
	Transfer Learning	MRI, CT scans	F1-score, precision, recall	[77]
	Generative adversarial networks (GANs)	X-ray images	AUC, dice coefficient	[78]
Monitoring patient vital signs and activity levels	Wearable Sensors	Patient health records	Accuracy, precision, recall	[79]
	Internet-of-Things (IoT) devices	Wearable devices data	Correlation, error rates	[80]
	Machine learning models	Continuous monitoring data	Sensitivity, specificity	[81]
Drug discovery and development through AI-driven research	Deep learning models	Chemical databases	ROC AUC, F1-score	[82]
	Reinforcement learning	Molecular structures	Binding affinity, accuracy	[83]
	Natural language processing (NLP)	Biomedical literature	Precision, recall, F-measure	[84]

15.12 Innovative Solutions for Retail and Consumer Goods

Cutting-edge technologies such as CV, ML, and robotics are disrupting traditional sales and consumption. Shelf tracking and inventory management can benefit from CV-based product recognition. Algorithms embedded in RFID technology improve accuracy, precision, and recall [94, 95]. Additionally, by using sales data and historical inventory to reduce MAPE and RMSE metrics, use ML models to predict demand forecasting and inventory optimization Robotics and automation system variety plays an important role in shelf-filling and inventory management, improving efficiency and reducing warehouse costs retail environments Becomes [96]. Moreover, personalized marketing customer behavior analytics are based on DL algorithms for facial expression recognition and sentiment analysis, facilitating sentiment analysis and sentiment detection for tracking customer interactions and content Customer purchase history and browsing behavior [97, 98]. Data mining and predictive analytics CRM systems e-marketing and consumption patterns enable customers to understand lifetime value and potential purchases [99].

Table 15.8 Application areas in agriculture and environmental monitoring.

Application	Methodology	Dataset	Reference
Crop monitoring and yield prediction	Deep learning models for image analysis	UAV imagery, satellite imagery	[85]
	Machine learning algorithms for crop growth prediction	Crop field databases	[86]
	Remote sensing techniques for crop health assessment	Hyperspectral imaging data	[87]
Pest detection and targeted pesticide application	Computer vision for pest identification and tracking	High-resolution images of crop fields	[88]
	IoT sensors for real-time pest monitoring and alert systems	Field sensor data	[89]
	Deep reinforcement learning for optimizing pesticide application	Historical pest infestation data	[90]
Environmental monitoring for Pollution detection and conservation efforts	Convolutional neural networks for pollution identification	Environmental satellite imagery	[91]
	Sensor networks for water quality monitoring	River and lake sensor data	[92]
	Geographic information systems (GIS) for habitat conservation	Biodiversity databases	[93]

In addition, NLP techniques for sentiment analysis and customer feedback processing along with facial recognition technologies for security

and customer recognition using CNNs enabled, times to extract valuable insights from social media platforms and customer research [100]. These advances in multifactor authentication and privacy-preserving techniques highlight the advances in AI and CV in the retail industry enhancing operational efficiency and customer experience [101, 102]. Table 15.9 presents the methods, data types, and performance measures used in retail and consumer goods.

15.13 Revolutionizing Transportation and Logistics with AI and CV

Innovative applications of AI and CV technologies in transport and dynamic logistics are reinventing traditional practices, as shown in Table 15.10. Automobiles, from cities' internal vehicles to autonomous public busses, use data sources such as the Waymo Open Dataset and Apollo Scape [103–105]. Automated package sorting systems and last-mile delivery robots improve package sorting and delivery optimization by using USPS package statistics and Amazon fulfillment databases to increase efficiency and delivery time [106–108]. Traffic management forecasts for streamlining use intelligent traffic management and AI-based traffic light optimization systems to reduce accidents and travel times [109, 110]. In addition, historical traffic forecasts can be accurately predicted by analyzing weather and event data [111]. These developments highlight the transformational impact of AI and CV in optimizing transport services and increasing safety, efficiency, and accessibility in urban areas.

Table 15.9 Application areas within retail and consumer goods.

Application area	Method	Dataset	Performance metrics	References
Shelf monitoring and inventory management	CV-based object detection algorithms combined with RFID technology	Store shelves and inventory databases	Accuracy, precision, recall	[94]
	ML models for demand forecasting and inventory optimization	Sales data, historical inventory records	MAPE (mean absolute percentage error), RMSE	[95]
	Robotics and automation systems for shelf restocking and inventory management	Warehouse and store environments	Efficiency, cost reduction	[96]
Customer Behavior Analysis for Personalized Marketing	DL algorithms for facial expression recognition and emotion analysis	Customer interactions, surveillance footage	Sentiment analysis, emotion detection	[97]
	Data mining and predictive analytics on customer purchase history and browsing behavior	CRM systems, e-commerce platforms	Customer lifetime value, purchase probability	[98]
	NLP for sentiment analysis and customer feedback processing	Social media platforms, customer reviews	Sentiment analysis, text mining	[99]

Application area	Method	Dataset	Performance metrics	References
Facial recognition for security and customer identification	CNNs for facial recognition and biometric authentication	Security cameras, customer databases	False acceptance rate (FAR), false rejection rate (FRR)	[100]
	Multi-factor authentication systems integrating facial recognition with other biometrics	Access control systems, identity verification	Accuracy, security level	[101]
	Privacy-preserving facial recognition techniques using federated learning	Edge devices, decentralized networks	Privacy, accuracy, model aggregation	[102]

Table 15.10 Application areas within transportation and logistics.

Category	Application	Dataset	Performance metrics	Reference
Autonomous vehicles	Self-driving cars for urban transportation	Waymo Open Dataset, ApolloScape, KITTI	Accuracy, safety, speed	[103]
	Autonomous shuttles for public transportation	Transdev Autonomous Shuttles, Navya Autonomous Shuttles	Safety, accessibility	[104]
	Autonomous trucks for freight transportation	Embark, Waymo Via, TuSimple	Fuel efficiency, safety	[105]
Package sorting & delivery optimization	Automated parcel sorting systems	USPS Parcel Data, Amazon Fulfillment Centers	Efficiency, accuracy	[106]
	Last-mile delivery robots	Starship Robots, Amazon Scout, FedEx SameDay Bot	Delivery time, cost	[107]
	Drone delivery systems	UPS Flight Forward, Amazon Prime Air, Wing	Speed, accessibility	[108]
Traffic management & congestion prediction	Intelligent traffic control systems	Traffic cameras, GPS data, city traffic data	Congestion reduction, travel time improvement	[109]
	AI-based traffic signal optimization	Intersection cameras, urban traffic flow data	Wait times, throughput	[110]

Category	Application	Dataset	Performance metrics	Reference
	Predictive analytics for traffic flow	Historical traffic data, weather data, event data	Accurate predictions	[111]

15.14 Advancing AI through Case-Based Reasoning (CBR)

CBR in cognitive science and AI challenges the traditional view of meaning as established only in memory. Instead, CBR emphasizes the intelligent use of experiences for new, unsolved problems. The diagnosis of new patients is an invaluable tool in areas such as medical education, patient history, and treatment [112]. Figure 15.7 shows the basic framework of information-based design, which is a model for human reasoning and an effective tool for promoting AI development.

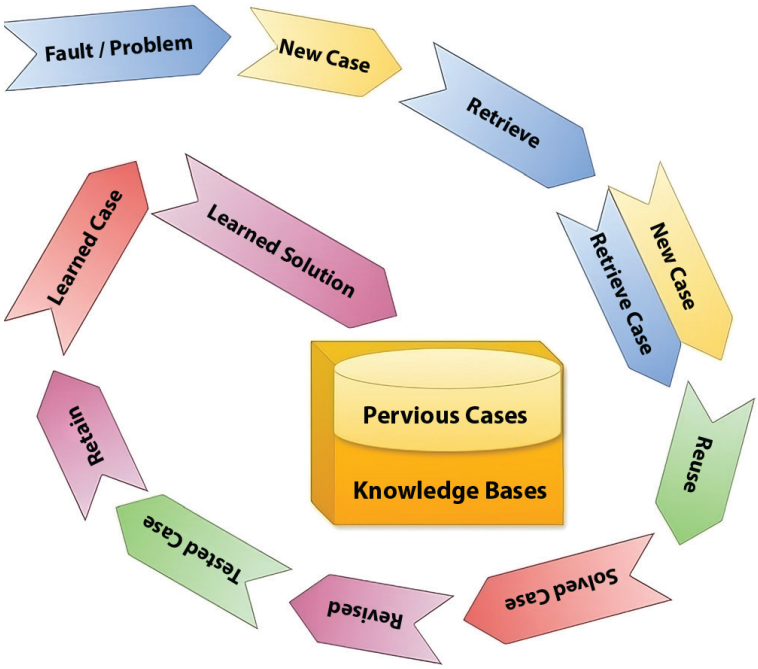


Figure 15.7 Basic structure of case-based reasoning for AI advancement.

Table 15.11 AI application in problem-solving.

Citation	Facts & figures	Examples
[112]	CBR involves gathering cases from similar experiences and tailoring their solutions to new challenges.	In medical education, case histories of patients and treatments are used to diagnose and treat new patients.
[113]	The visual case-based approach enhances AI in colon cancer treatment.	Kumar V. *et al.* proposed a visual casebased approach to improve AI in colon cancer treatment.
[114]	CBR is applied in error detection systems for drip production.	Khosravani *et al.* suggested employing CBR in error detection systems for drip production.
[115]	CBR-based data cleaning approach is introduced for classification and regression tasks.	Corrales *et al.* introduced a CBR-based data-cleaning approach for classification and regression tasks.

CBR finds applications in a wide range of industries. Kumar V. *et al.* [113] proposed a visual narrative-based approach to enhance AI in colon cancer screening. Khosravani *et al.* [114] proposed the benefits of CBR in point formation and fault detection systems. Furthermore, Corrales *et al.* [115] provided a CBR-based data-correction approach for classification and regression tasks. Table 15.11 shows the application of AI-based problem-solving. As the volume of data collected, increases, CBR systems are smarter because the time required to identify and process relevant data can increase, affecting system efficiency.

15.15 Text Mining and NLP in IAS

Text mining, also known as text data mining, uses techniques such as text mining, lexical analysis, pattern recognition, and data mining to analyze textual content [116]. NLP helps in mining information by processing human language for understanding and analysis [117]. NLP techniques include speech recognition, tokenization, lemmatization, and part-of-speech marking. The authors of [118] improved communication analysis in tweets by using multiple models while using weighted terms of deep neural networks for product sentiment inference. Figure 15.8 presents a developed semantic analysis in CV for social support robots and broader areas of great impact by AI.

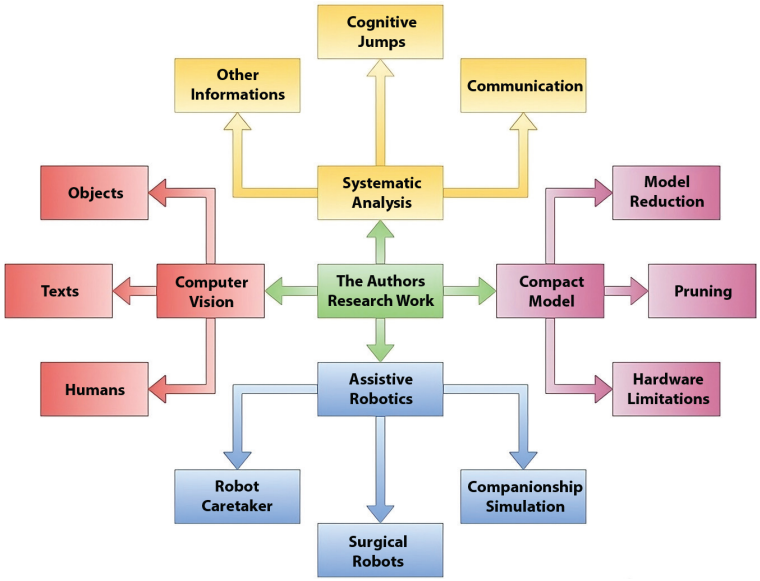


Figure 15.8 Semantic analysis of computer vision in socially helpful robotics review.

Similarly, [119] studied emotional expressions in tweets to assess the security and authenticity of events. Other researchers are exploring the use of social media data for health communication. ML technologies, and DL models such as RNNs and CNNs, excel in automating and intelligently extracting meaning from large data sets [120]. Pattern recognition underlies AI systems, enabling supervised and unsupervised modeling recognition in data heads, and combining analysis and optimization techniques plays a key role in developing models to solve real-world problems. AI materials have proven effective in addressing the challenges in healthcare, cybersecurity, entrepreneurship, social networks, and robotics [121]. Table 15.12 presents the AI and CV applications in the analysis of information.

AI is shaping almost every aspect of business and personal life worldwide. The “Handbook of Intelligent Automation Systems using Computer Vision and AI” explores the use of AI to achieve technological advances in areas such as automation, healthcare, agriculture, the IoT, and computing. The safety of AI plays a key role in enabling automation by making intelligent decisions that enhance the reliability, efficiency, and speed of tasks traditionally performed by humans. The use of virtual assistants in customer service departments not only reduces costs but also enhances the overall customer experience. An overview of AI techniques in real-world scenarios is presented in Table 15.13. With the rapid development of AI, intelligence-based automation is expected to become an essential tool for companies and individuals worldwide.

Table 15.12 AI and CV applications in information analysis.

Citation	Facts & figures	Examples
[116]	NLP enables machines to understand and analyze human language.	NLP techniques encompass tasks like speech recognition, tokenization, sentiment analysis, and chatbot development.
[117]	Enhance sentiment analysis in tweets using an ensemble of models.	Phan *et al.* improved sentiment analysis in tweets using an ensemble of models.
[118]	Employ weighted word embeddings in deep neural networks to evaluate product emotions.	Onan *et al.* used weighted word embeddings in deep neural networks for product emotion evaluation.
[119]	Investigate sentiments expressed in tweets to assess event safety and veracity.	Mahmood *et al.* studied sentiments in tweets to assess event safety and veracity.
[120]	Social media data is used for health communication.	Social media data is utilized for health communication
[121]	AI has made significant impacts across diverse fields such as health, cybersecurity, business, social networks, and robotics.	AI has demonstrated efficacy in various fields including health, cybersecurity, and robotics.

Table 15.13 Overview of AI techniques and applications in real-world scenarios.

AI techniques	Application areas	Tasks	Complexity (1–10)	Number of citations	References
Machine learning	Healthcare	Recommendation systems	7	12	[12]
	Cybersecurity	Anomaly and Attack Detection	9	36	[73, 76, 78, 89]
	Smart city	Smart parking pricing system	5	8	[109]
		Hotel recommendation	6	10	[105]
Neural network and deep learning	Healthcare	Diagnosis of COVID-19	8	20	[9, 41]
	Cybersecurity	Malware detection	9	24	[48, 100]
	Smart city	Smart parking system	7	15	[69]
	Smart agriculture	Plant disease detection	8	18	[6]
	Business and finance	Stock trend prediction	8	30	[8, 22, 54]
		An intelligent chatbot	7	14	[28]
	Visual recognition	Facial expression analysis	6	10	[127]

AI techniques	Application areas	Tasks	Complexity (1–10)	Number of citations	References
Data mining, knowledge discovery, and advanced analytics	Education	Decision support systems	7	15	[35]
	Business	Maximizing competitive advantage	8	22	[5]
	Cybersecurity	Human-centered data mining	9	28	[1]
	Diagnostic analytics	To mature gas fields	7	12	[70]
	Prescriptive analytics	Optimizing outpatient appointment	8	18	[94]
Rule-based modeling and decision-making	Intelligent systems	Mining contextual rules	9	25	[88]
	Healthcare	Identifying risk factors	8	20	[13]
	Recommendation system	Web page recommendation	6	10	[11, 103]
	Smart systems	Risk prediction	7	15	[60]
Fuzzy logic–based approach	Healthcare	Heart disease diagnosis	8	18	[74]
	Agriculture	Smart irrigation	7	14	[51]
	Cybersecurity	Network anomaly detection system	9	24	[34]
	Business	Customer satisfaction	8	20	[44]

AI techniques	Application areas	Tasks	Complexity (1–10)	Number of citations	References
Knowledge representation, uncertainty reasoning, and expert system modeling	Smart Systems	Smart traffic monitoring	8	22	[29]
	Cloud computing cybersecurity	Ontology data access control	7	16	[49]
	Mobile expert system	Vulnerability management	8	20	[97]
		Personalized decision-making	7	15	[90]
		Personalized decision-making	7	15	[90]
Case-based Reasoning	Healthcare	Breast cancer management	9	24	[52]
	Smart city	Energy management	7	15	[30]
	Smart industry	Fault detection system	8	20	[47]
	Recommendation systems	Classification and regression tasks	7	18	[17]
Text mining and natural language processing	Sentiment analysis	Sentiment analysis of tweets	8	20	[68]
	Business	Product reviews sentiment	7	16	[62]
	Cybersecurity	Estimating the security of events	8	20	[96]

AI techniques	Application areas	Tasks	Complexity (1–10)	Number of citations	References
	Healthcare	Effectiveness of social media	7	14	[60]
Visual Analytics, Computer Vision, and Pattern Recognition	Healthcare	Cervical cancer diagnostics	9	24	[25]
	Computer vision	Human fall detection	8	20	[38]
	Visual analytics	Navigation mark classification	7	15	[63]
Hybrid Approach, Searching, and Optimization	Mobile application	Personalized decision-making	7	16	[90]
	Recommendation systems	Personalized hotel recommendation	8	20	[73]
	Sentiment analysis	Tweet sentiment accuracy analysis	7	18	[68]
	Business	Customer satisfaction	8	20	[44]
	Cybersecurity	Optimum feature selection	9	24	[28, 61]

- AI is set to transform all aspects of business and personal life globally.
- AI is revolutionizing industrial engineering, medical products, agriculture, the Internet, and cybersecurity.
- Reducing human interference is a major focus of current AI efforts.
- Automation streamlines IT operations by implementing sequential processes, scripts, or actions.
- AI enhances automation by enabling reliability, effort, and speed to make intelligent decisions in tasks traditionally performed by humans.
- Adding virtual assistants to customer service departments reduces costs and enhances the customer experience.
- AI-powered automation will be widespread, affecting almost every business and individual worldwide.

Applications, technologies, and measurement capabilities for various industries are shown in Table 15.14. In the manufacturing sector, this includes CAD/CAM automated inspection and ML using images from vision systems used to test models for greater accuracy and error detection, as well as IoT for predictive maintenance and DB deployment. Assembly is the process of assembling product parts using robotic computer vision. Management RFID is a method of identifying stock stored using computer vision. The supply chain automation and big data are used to increase the resource utilisation, efficiency of the automation systems. Further, ML, robotics, IoT, Blockchain, CNN, and NLP are used to increase accuracy, specificity, and recall of the automation systems. In this context, called "retail," CV and RFID are used for shelves, DL and NLP to track customer behavior, CNN and MFA for facial recognition and leverage accuracy, translation, and security in sales Automation technology in transportation, AI in autonomous vehicles, vision, Robotics and AI in package sorting, and the integration of intelligent cameras and GPS for traffic management, safety, and speed management at the knowledge level, probabilistic reasoning, and case include -based -Cross -domain methods such as reasoning, text mining/NLP and improve reasoning ability, decision accuracy, effective problem solving, and information extraction respectively.

Table 15.14 Summary of applications, strategies, technologies, and metrics in various industries.

Domain	Application	Techniques	Technologies	Metrics
Manufacturing	Quality inspection	Automated inspection	Computer vision, ML	Accuracy, detection rate
	Predictive maintenance	Predicting failures	IoT, data analytics	Downtime, maintenance cost
	Robotic assembly	Automated assembly	Robotics, computer vision	Efficiency, precision
	Inventory management	Tracking and management	RFID, computer vision	Turnover, accuracy
	Process optimization	Analyzing processes	Data analytics, ML	Utilization, efficiency
	Supply chain automation	Integrating processes	Robotic, IoT, blockchain	Transparency, efficiency
Healthcare	Medical image analysis	CNN, transfer learning	Deep learning	Accuracy, specificity
	Patient monitoring	Wearable sensors, IoT	Sensor data	Precision, recall
	Drug discovery	Deep learning, NLP	Chemical databases	ROC AUC, F1-score
Agriculture	Crop monitoring	ML, remote sensing	UAV, satellite imagery	Yield accuracy, health
	Pest detection	Computer vision, IoT	High-res images, sensors	Detection rate, optimization
	Environmental monitoring	CNN, sensor networks	Satellite imagery, GIS	Pollution detection
Retail	Shelf monitoring	Computer vision, RFID	Store shelves data	Accuracy, precision
	Customer behavior analysis	Deep learning, NLP	CRM systems, social media	Sentiment, emotion detection
	Facial recognition	CNNs, MFA	Security cameras	False acceptance/rejection

Domain	Application	Techniques	Technologies	Metrics
Transportation	Autonomous vehicles	Self-driving tech	AI planning, vision	Safety, speed
	Package sorting/delivery	Sorting, last-mile delivery	Robotic, AI	Efficiency, delivery time
	Traffic management	Intelligent control	Traffic cameras, GPS	Congestion, travel time
Cross-Domain Techniques	Knowledge representation	Ontologies	Rule-based systems	Reasoning capabilities
	Probabilistic reasoning	Bayesian, Markov models	Logic, probability theory	Decision accuracy
	Case-based reasoning	Past experiences	CBR systems	Problem-solving efficiency
	Text mining and NLP	Lexical analysis, pattern	Deep learning, NLP	Sentiment, information extraction

15.16 Exploring Artificial Intelligence Applications and Challenges

Computationally intelligence includes a wide range of techniques aimed at extracting insights and knowledge from data, including data processing, ML, pattern recognition, and NLP. These technologies can be used roles in various sectors like; transportation, infrastructure, agriculture, finance, and defense contributing to the development of autonomous systems [130]. These interdisciplinary efforts include the engineering approaches, required to develop intelligent mixing systems for A-based technologies. The ultimate goal is to have an intelligent system, capable of analyzing and delivering information efficiently.

Industry 4.0, also known as the fourth technological revolution, emphasizes the importance of automation, intelligence, and information systems across industries. Industries such as chatbots, customer self-service, payment automation, and the detection of cybersecurity threats are examples of this. AI and ML are performed in transforming business operations, processes, cybersecurity, and financial security [131]. However, there are notable challenges in adapting algorithms and models to different data types and real-world situations.

Efforts to solve real-world problems with AI require identifying the most appropriate technologies for given problems and designing effective systems to address the associated challenges This requires building intelligent systems that can sense and interpret their environment, oscillating with decision uncertainty, and small models for the IoT [132]. Moreover, ensuring proper data integration and preprocessing and reducing risk and uncertainty are important elements in implementing AI solutions.

Intelligent technologies have tremendous potential to boost economic growth and solve complex challenges in various industries. However, leveraging this potential requires significant efforts to develop new solutions, refine existing technologies, and implement supporting systems. With the right skills and credentials, AI professionals can open doors to career opportunities, and play a critical role in shaping the future of technology and society [133, 134].

15.17 Exploring Artificial Intelligence in Computer Vision Tasks

Various applications and data types in computer vision processing play an important role in the development and implementation of the technology, as shown in the table above Traffic lights, traffic signals, pedestrians, vehicle detection, and so on Object detection services by COCO, CityScape, CIFAR-10/CIFAR-100 and mapping, precision, and accuracy [122]. Performance metrics implementation classification services including speed estimation and tracking methods use COCO, CityScape, and BD100K data systems with land imagery as a performance metric [123]. Image enhancement services such as rainfall and ultra-high resolution rely on REDS and PSNR as performance metrics [124]. Search engines such as pedestrian and vehicle detection use data such as MOT19 and MOTA as performance indicators [125]. Incident and prediction applications such as accident detection and accident analysis use data such as UCF101 and Kinetics600 accuracy and mAP as performance metrics [126] as anomaly detection applications, AUC and accuracy data with abnormal behavior and vehicle including road violation detection UCSD uses Ped1 and UMN as performance metrics [127]. Density analysis projects, including pedestrian and vehicle density analysis, are based on datasets such as Oxford 5K and UCSD Mall with MAE and MSE performance metrics included [128]. Image and event recovery applications such as target disk and digital archive retrieval rely on datasets such as Oxford 5K and Pascal VOC to measure performance [129, 130]. Table 15.15 lists applications and datasets in computer vision tasks [131, 132]. Computer vision in solving real-world challenges, and improving technical capabilities is emphasized through these applications and datasets.

Table 15.15 Applications and datasets in computer vision tasks.

Computer vision function	Application areas	Sample datasets	Performance metrics
Object detection [122]	Traffic lights	COCO, CityScape	mAP, Accuracy, Precision, Recall
	Traffic signs	ImageNet, LISA, Pascal VOC	AP, accuracy, precision
	Pedestrian detection	CIFAR-10/CIFAR-100	mAP, accuracy
	Vehicle type/counting	GTSDB, Pascal VOC	Accuracy, AP
Object segmentation [123]	Speed estimation	COCO, CityScape, KITTI	mAP
	Road line tracking, Route optimization	BD100K, LISA	
Image enhancement [124]	Raindrop removal, super-resolution, image sharpening	REDS	PSNR
Object tracking [125, 135]	Fish-eye to Cartesian, pedestrian/vehicle, speed detection, route extraction	MOT19	MOTA
Event identification/prediction [126]	Accident recognition/prediction, congestion estimation, dangerous situations	UCF101, Kinetics600	Accuracy, mAP
Anomaly detection [127, 137]	Abnormal behavior, traffic rule violations	UCSD Ped1, UCSD Ped 2, Avenue, UMN, UCF Real World	AUC, accuracy, mAP
Density analysis [128]	Pedestrian/vehicle, tensity analysis	Oxford 5K, UCSD Mall, ShanghaiTech, UCF_CC_50, WorldExpo'10	MAE, MSE
Image/event search [129, 136]	Target plate search, digital archives	Oxford 5K, Pascal VOC	Accuracy

Computer vision function	Application areas	Sample datasets	Performance metrics
	search, object detection		

15.18 Conclusion

The intersection of AI and CV signals profound changes across industries, transforming traditional processes and increasing productivity delivered by supervised, unsupervised, or hybrid AI innovations and processes are developed using rule-based systems and ambiguous logic, resulting in manufacturing, healthcare, agriculture, retail, and other industries can make intelligent decisions. AI and CV solutions in transportation and logistics enhance road optimization, traffic management, safety protocols, operational analysis streamlining, and spectator experience enhancement. The advancement of AI and CV technologies in crop management, monitoring, and secure transportation systems is paving the way for industry transformation and global living improvement. Integrating AI and continuous improvement is reshaping industry norms and engaging with the world innovatively.

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Note

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